```
Experiment 5: 90-degree clockwise obstacle rotation
Obstacle count (before/after): 57 57
Environment check
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Grid: 20 x 20
Start: (0, 0)
Goal: (19, 19)
Obstacles: 57
Shortest valid path: 38 steps

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EXPERIMENT 5 STARTED
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Training: Sparse Q-Learning
 Seed 1/10 completed in 13.7 sec
 Seed 2/10 completed in 6.8 sec
 Seed 3/10 completed in 8.5 sec
 Seed 4/10 completed in 10.4 sec
 Seed 5/10 completed in 9.4 sec
 Seed 6/10 completed in 14.3 sec
 Seed 7/10 completed in 5.5 sec
 Seed 8/10 completed in 8.6 sec
 Seed 9/10 completed in 12.6 sec
 Seed 10/10 completed in 7.7 sec
Sparse Q-Learning completed in 97.6 seconds.

Training: Fixed PBRS Manhattan
 Seed 1/10 completed in 61.0 sec
 Seed 2/10 completed in 58.1 sec
 Seed 3/10 completed in 58.7 sec
 Seed 4/10 completed in 57.1 sec
 Seed 5/10 completed in 56.9 sec
 Seed 6/10 completed in 58.2 sec
 Seed 7/10 completed in 57.0 sec
 Seed 8/10 completed in 57.4 sec
 Seed 9/10 completed in 57.9 sec
 Seed 10/10 completed in 57.5 sec
Fixed PBRS Manhattan completed in 579.6 seconds.

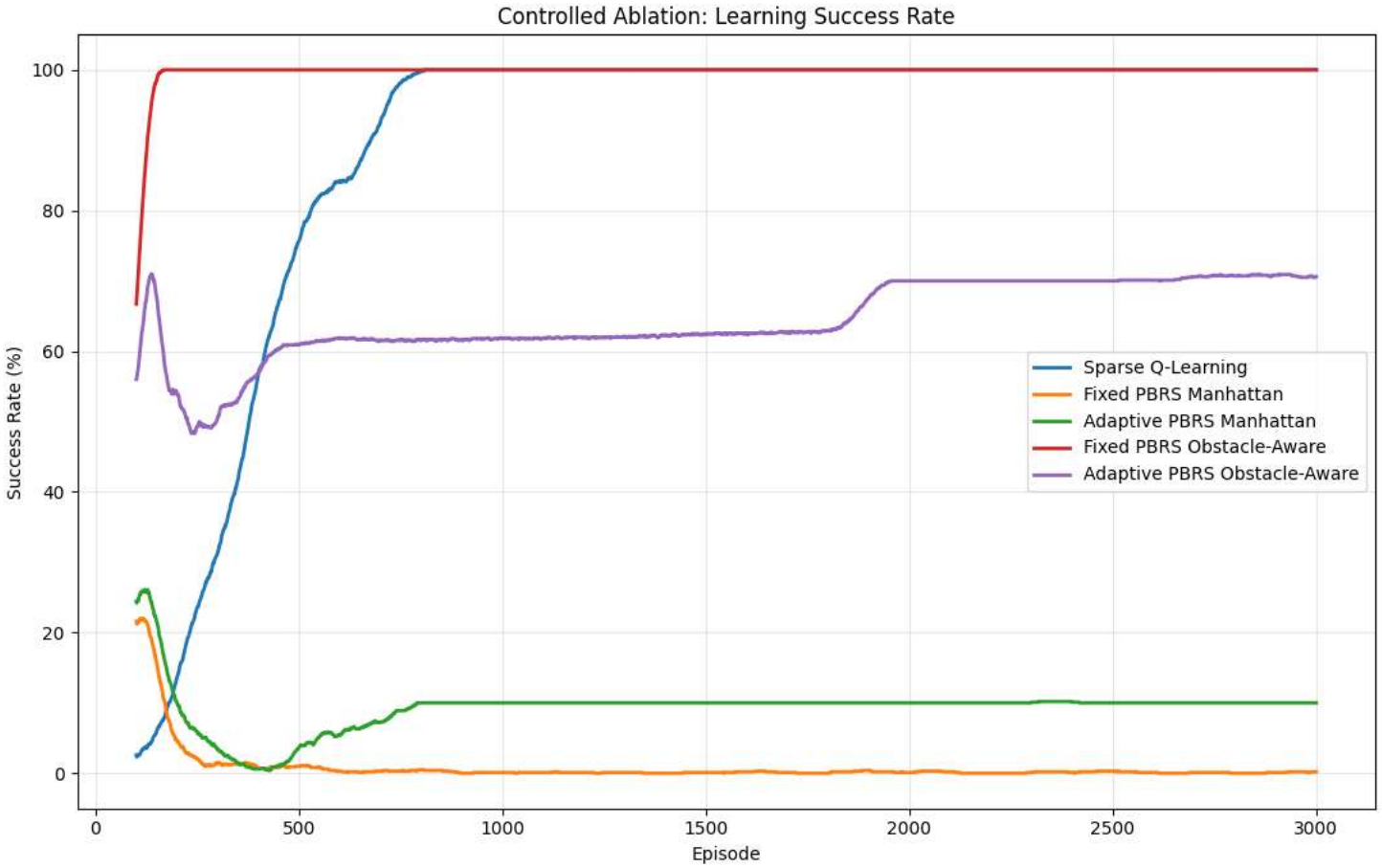
Training: Adaptive PBRS Manhattan
 Seed 1/10 completed in 59.0 sec
 Seed 2/10 completed in 58.7 sec
 Seed 3/10 completed in 13.9 sec
 Seed 4/10 completed in 83.6 sec
 Seed 5/10 completed in 62.3 sec
 Seed 6/10 completed in 59.3 sec
 Seed 7/10 completed in 58.5 sec
 Seed 8/10 completed in 57.6 sec
 Seed 9/10 completed in 59.0 sec
 Seed 10/10 completed in 58.1 sec
Adaptive PBRS Manhattan completed in 569.9 seconds.

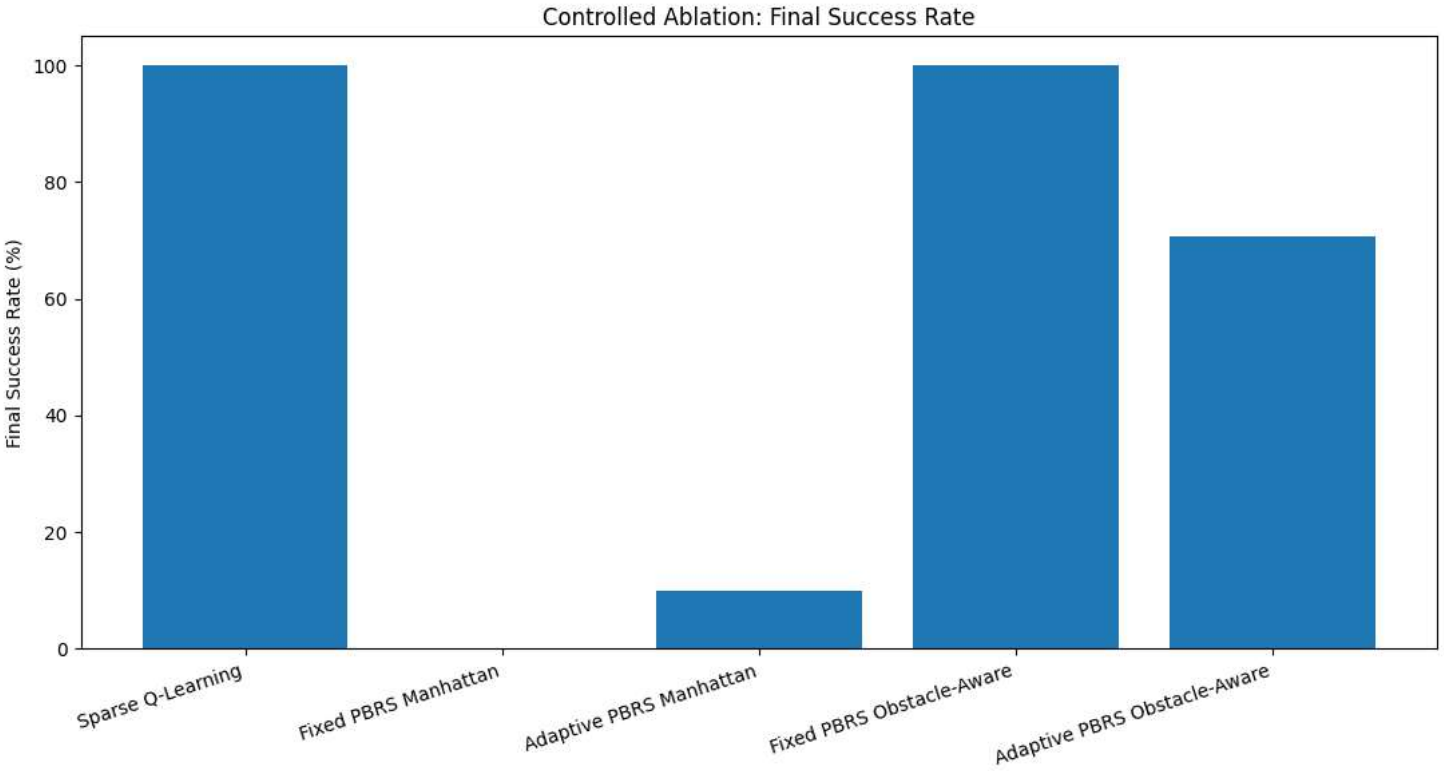
Training: Fixed PBRS Obstacle-Aware
 Seed 1/10 completed in 4.8 sec
 Seed 2/10 completed in 5.0 sec
 Seed 3/10 completed in 4.4 sec
 Seed 4/10 completed in 5.5 sec
 Seed 5/10 completed in 4.5 sec
 Seed 6/10 completed in 4.4 sec
 Seed 7/10 completed in 5.6 sec
 Seed 8/10 completed in 4.4 sec
 Seed 9/10 completed in 4.3 sec
 Seed 10/10 completed in 5.5 sec
Fixed PBRS Obstacle-Aware completed in 48.3 seconds.

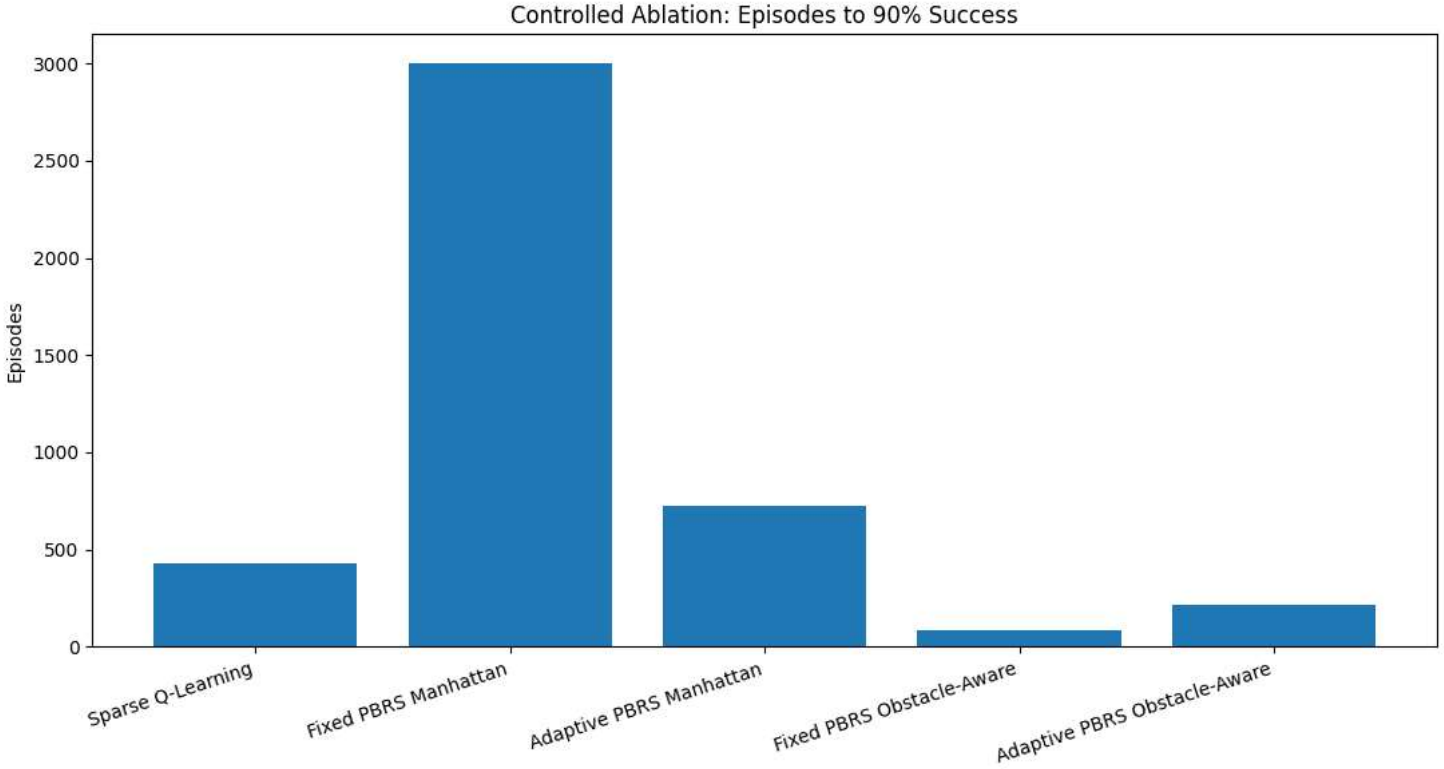
Training: Adaptive PBRS Obstacle-Aware
 Seed 1/10 completed in 58.2 sec
 Seed 2/10 completed in 6.2 sec
 Seed 3/10 completed in 31.5 sec
 Seed 4/10 completed in 7.0 sec
 Seed 5/10 completed in 5.3 sec
 Seed 6/10 completed in 6.4 sec
 Seed 7/10 completed in 6.6 sec
 Seed 8/10 completed in 59.0 sec
 Seed 9/10 completed in 59.0 sec
 Seed 10/10 completed in 5.6 sec
Adaptive PBRS Obstacle-Aware completed in 244.8 seconds.

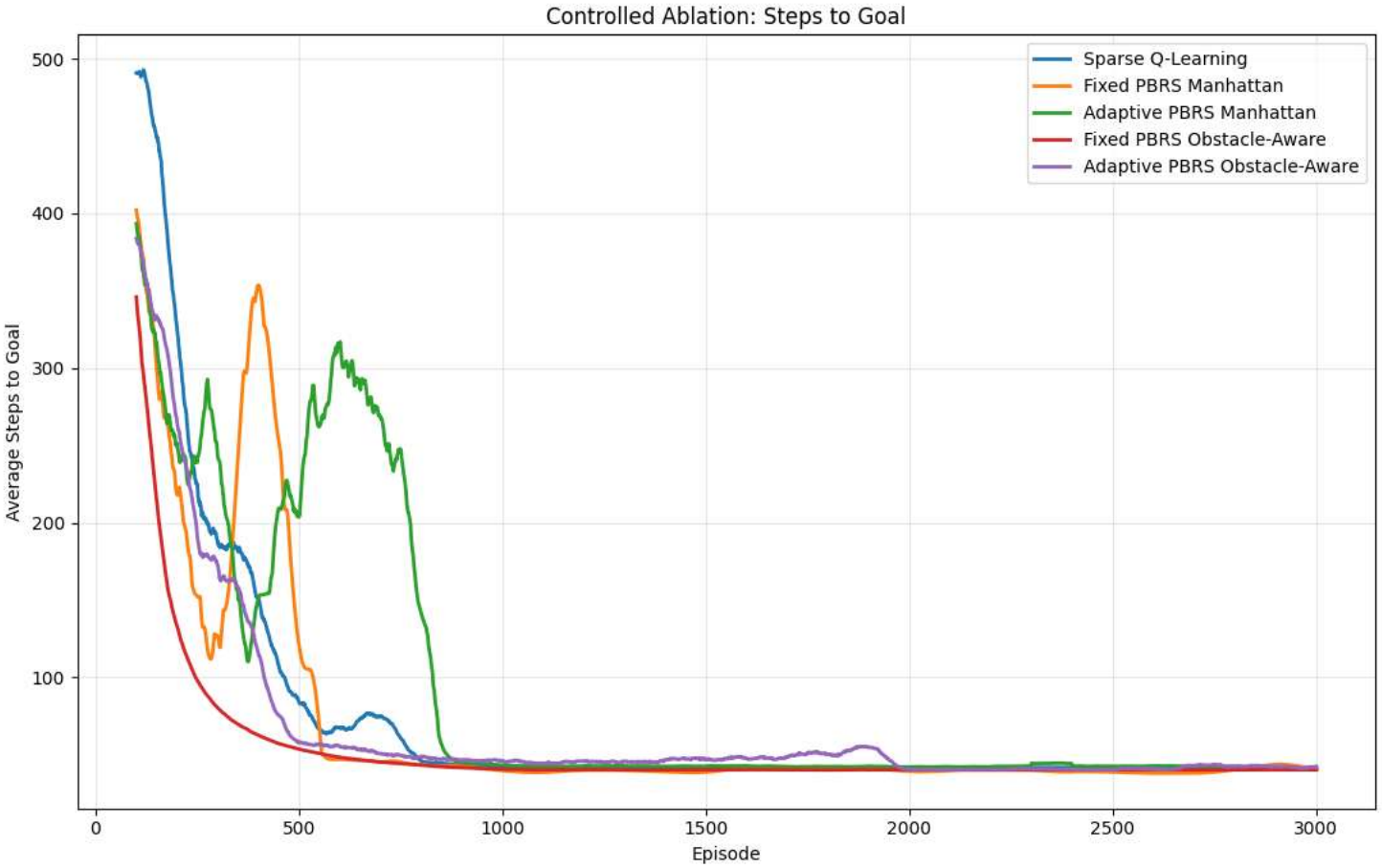
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ROTATED-OBSTACLE CONTROLLED ABLATION RESULTS
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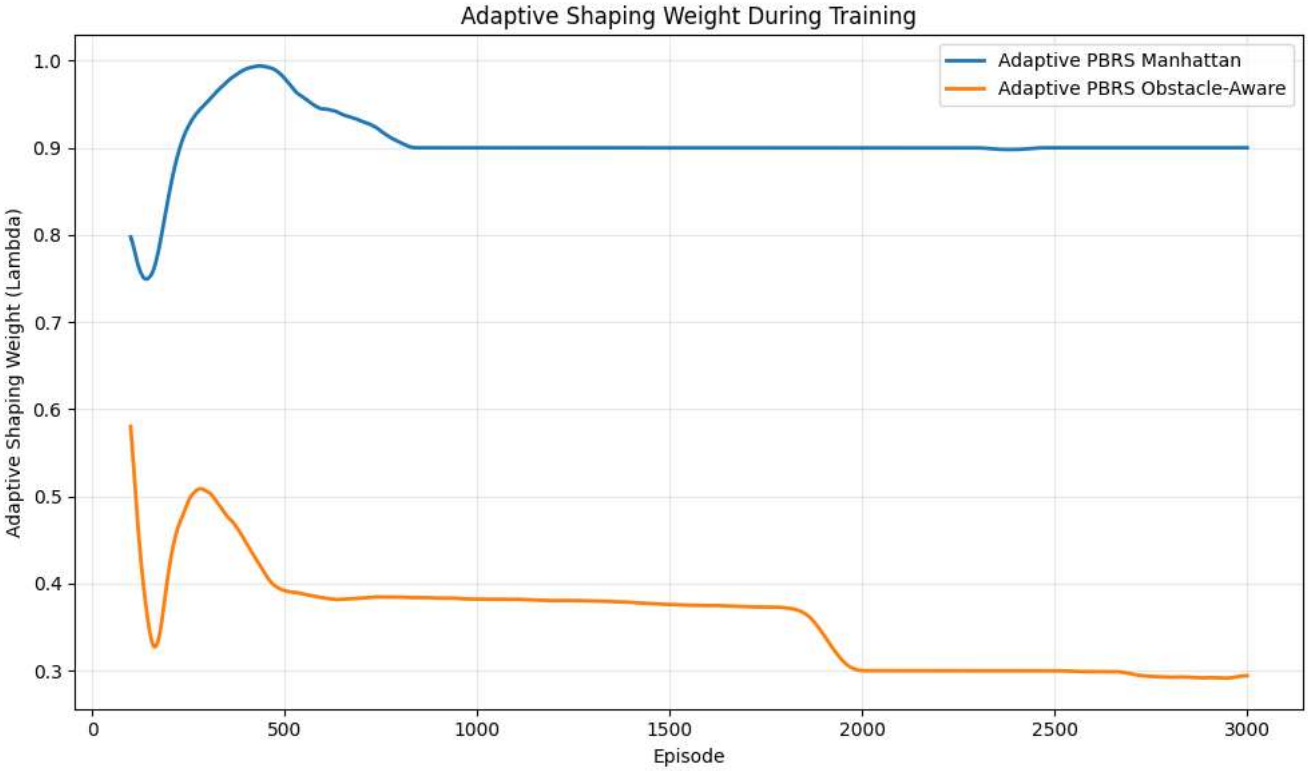
	Method	Final Success Rate (%)	Success SD	Average Steps to Goal	First Successful Episode	Seeds Reaching 90%	Episodes to 90% Success
0	Sparse Q-Learning	100.00	0.00	41.69	42.1	10	430.60
1	Fixed PBRS Manhattan	0.15	0.47	41.67	9.3	0	NaN
2	Adaptive PBRS Manhattan	10.00	31.62	40.65	9.3	1	727.00
3	Fixed PBRS Obstacle-Aware	100.00	0.00	40.25	9.1	10	86.10
4	Adaptive PBRS Obstacle-Aware	70.70	47.22	68.78	9.1	8	213.25

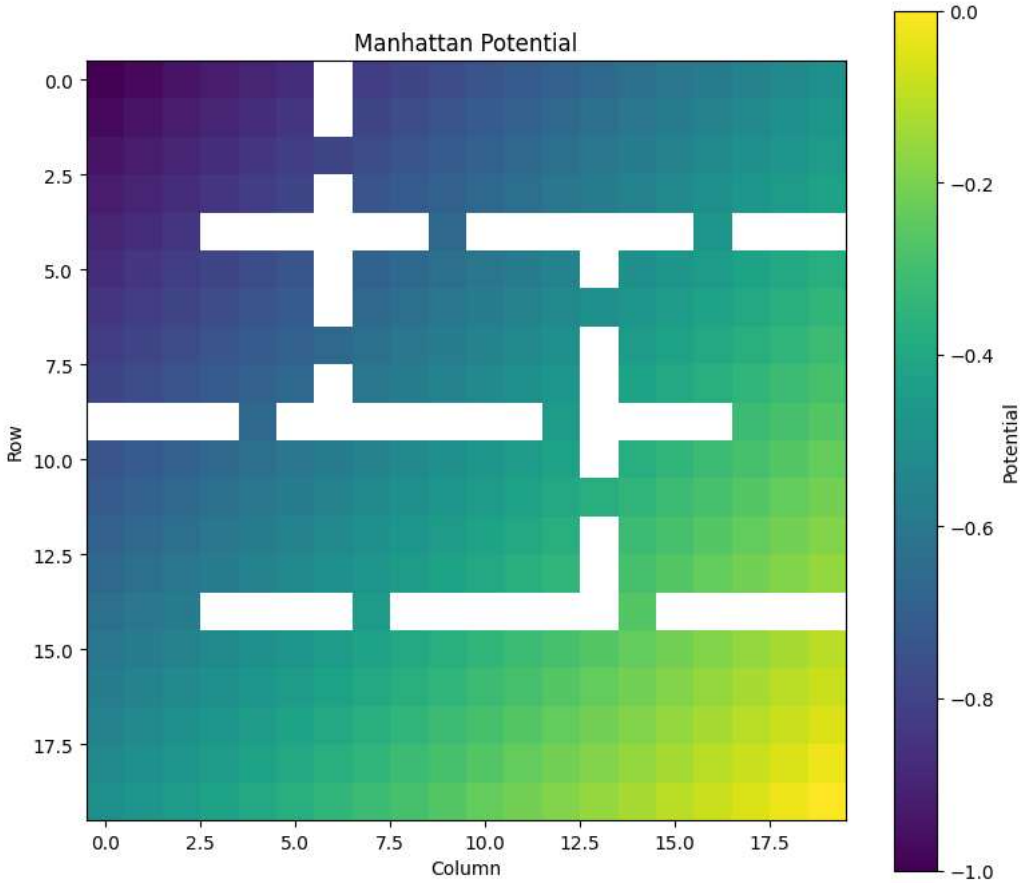


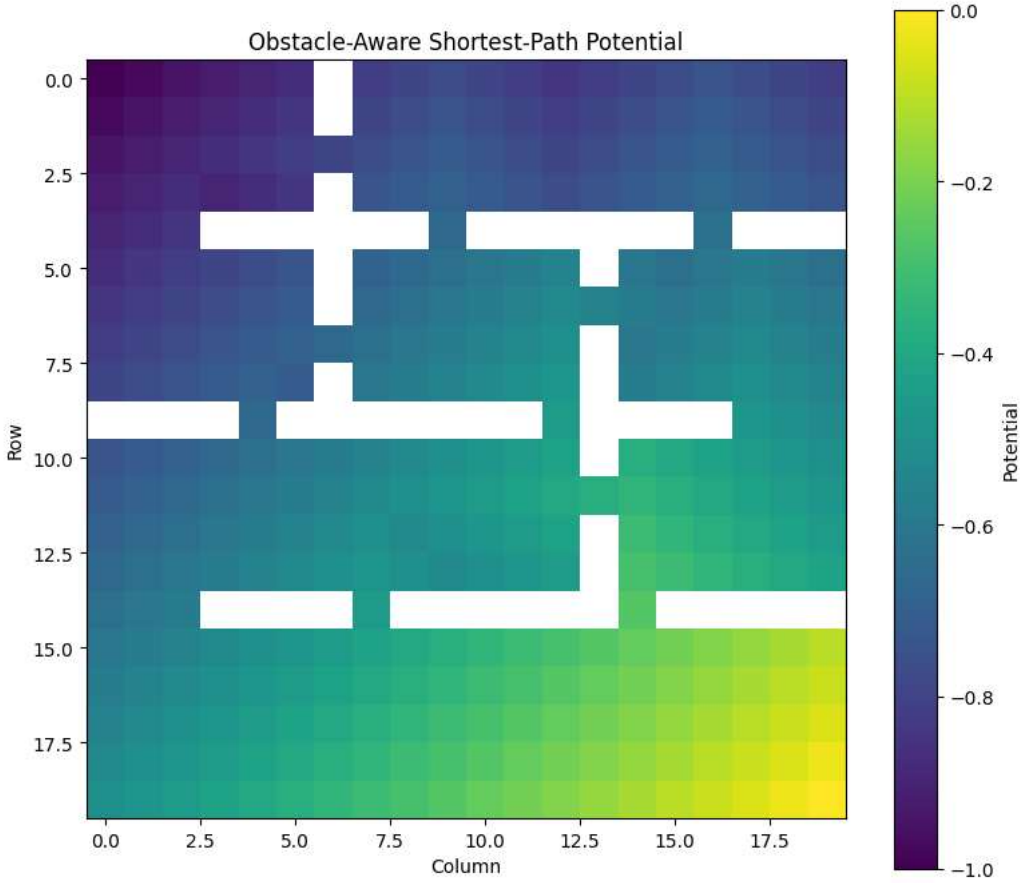

Controlled Ablation: Final Success Rate
Final Success Rate (%)
0
20
40
60
80
100
Sparse Q-Learning
Fixed PBRS Manhattan
Adaptive PBRS Manhattan
Fixed PBRS Obstacle-Aware
Adaptive PBRS Obstacle-Aware


Controlled Ablation: Episodes to 90% Success
Episodes
0
500
1000
1500
2000
2500
3000
Sparse Q-Learning
Fixed PBRS Manhattan
Adaptive PBRS Manhattan
Fixed PBRS Obstacle-Aware
Adaptive PBRS Obstacle-Aware


Controlled Ablation: Steps to Goal
Sparse Q-Learning
Fixed PBRS Manhattan
Adaptive PBRS Manhattan
Fixed PBRS Obstacle-Aware
Adaptive PBRS Obstacle-Aware
Average Steps to Goal
Episode
500
400
300
200
100
0
500
1000
1500
2000
2500
3000


Adaptive Shaping Weight During Training
Adaptive PBRS Manhattan
Adaptive PBRS Obstacle-Aware
Adaptive Shaping Weight (Lambda)
Episode
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0
500
1000
1500
2000
2500
3000






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================================================================================================
EXPERIMENT 5 COMPLETED
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Total execution time: 25.79 minutes

FINAL TABLE:
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Method	Final Success Rate (%)	Success SD	Average Steps to Goal	First Successful Episode	Seeds Reaching 90%	Episodes
Sparse Q-Learning	100.00	0.00	41.69	42.1	10	
Fixed PBRS Manhattan	0.15	0.47	41.67	9.3	0	
Adaptive PBRS Manhattan	10.00	31.62	40.65	9.3	1	
Fixed PBRS Obstacle-Aware	100.00	0.00	40.25	9.1	10	
Adaptive PBRS Obstacle-Aware	70.70	47.22	68.78	9.1	8	

Generated files:
1. experiment5_summary.csv
2. experiment5_seed_results.csv
3. exp5_success_rate.png
4. exp5_final_success.png
5. exp5_convergence.png
6. exp5_steps.png
7. exp5_lambda.png
8. exp5_manhattan_potential.png
9. exp5_obstacle_potential.png